

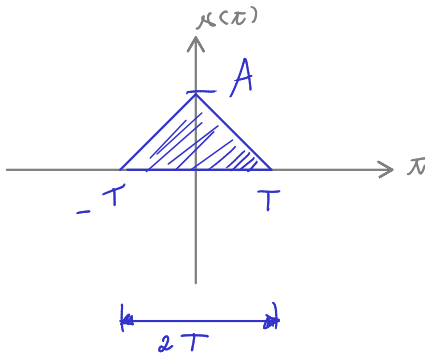
24 de agosto de 2026

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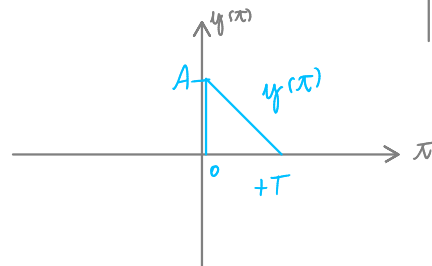
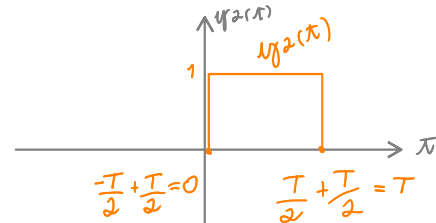
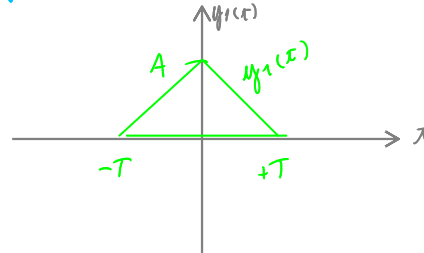
1. Represente graficamente os seguintes sinais:

a) $x(t) = A \operatorname{tri}\left(\frac{t}{T}\right)$;

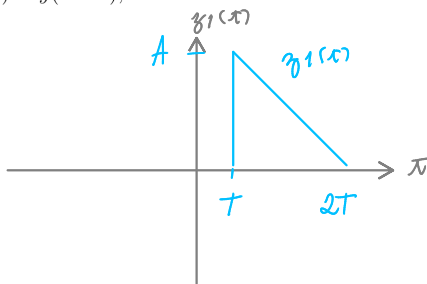


b) $y = A \operatorname{tri}\left(\frac{t}{T}\right) \operatorname{rect}\left(\frac{t}{T} - \frac{1}{2}\right) = A \operatorname{tri}\left(\frac{t}{T}\right) \operatorname{rect}\left(\frac{1}{T}\left[t - \frac{T}{2}\right]\right)$

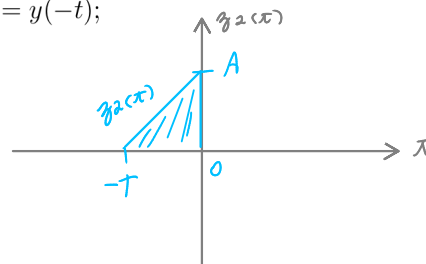
$y(t) = y_1 \cdot y_2$



c) $z_1(t) = y(t - T)$;

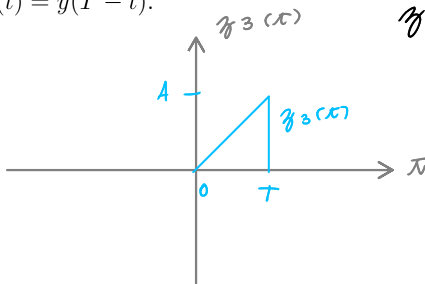
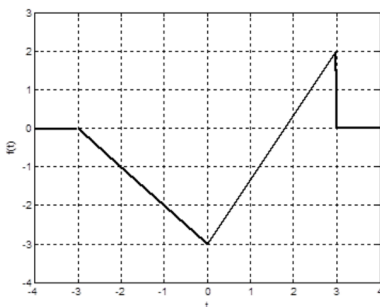


d) $z_2(t) = y(-t)$;

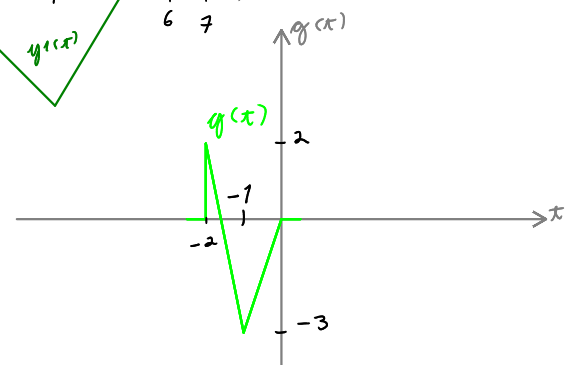
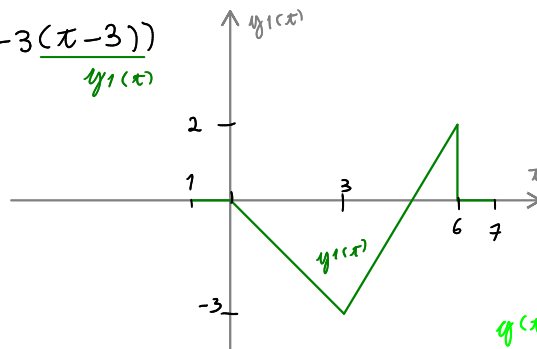


e) $z_3(t) = y(T - t)$.

$z_3(t) = y(-(t + T))$

2. Considere a função $f(t)$ mostrada na figura abaixo:

$g(t) = f(-3(t-3))$



a) Esboce a função $g(t) = f(9 - 3t)$;

b) Calcule a energia e a potência média de $f(t)$. Trata-se de um sinal de potência ou de energia?

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \underbrace{\int_{-T}^T |f(x)|^2 dx}_{\text{VALOR FINITO}}$$

$$\lim_{T \rightarrow \infty} \frac{V.F.}{2T} = 0 \therefore P = 0$$

$$E = \int_{-\infty}^{\infty} |x(x)|^2 dx = \int_{-3}^0 (-x+3)^2 dx + \int_0^3 \left(\frac{5}{3}x-3\right)^2 dx$$

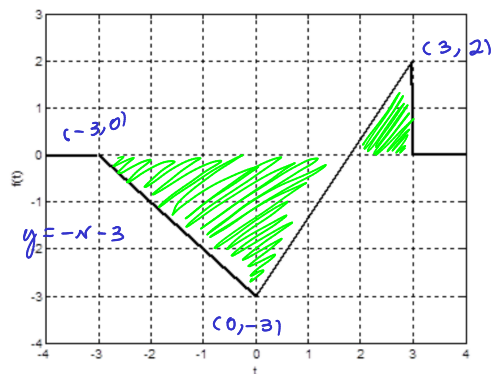
$$= \int_{-3}^0 x^2 + 6x + 9 dx + \int_0^3 \frac{25}{9}x^2 - 10x + 9 dx$$

$$= \left| \frac{x^3}{3} + 3x^2 + 9x \right|_{-3}^0 + \left| \frac{25x^3}{9 \cdot 3} - \frac{10x^2}{2} + 9x \right|_0^3$$

$$-(-9 + 27 - 27) + \left(\frac{25 \cdot 3^3}{3^2 \cdot 3} - 5 \cdot 3^2 + 3^2 \cdot 3 \right)$$

$$+ 9 + (25 + 3^2(-5+3)) = +9 + (25 - 3^2 \cdot 2)$$

$$= +9 + 25 - 18 = 16 \Rightarrow E = 16$$



$$ax + b = y$$

$$0 \cdot a + b = -3$$

$$b = -3$$

$$-3a + b = 0$$

$$0 \cdot a + b = -3 \therefore b = -3$$

$$-3a - 3 = 0 \therefore a = -1$$

$$3a - 3 = 2$$

$$3a = 5$$

$$a = 5/3$$

Sinal de energia pois a potência média é zero e possui valor finito maior que zero para E .

3. Determine se os seguintes sinais são sinais de potência ou sinais de energia.

Calcule a energia e a potência média de cada sinal.

a) $x_1(t) = \cos(\pi t) \sin(3\pi t)$

b) $x_2(t) = \begin{cases} \cos(3\pi t), & -3 \leq t < 3 \\ 0, & \text{fora;} \end{cases}$

c) $x_3(t) = \begin{cases} t, & 0 \leq t \leq 2 \\ 4-t, & 2 \leq t \leq 4 \\ 0, & \text{fora;} \end{cases}$

$$* \sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha - \beta) + \sin(\alpha + \beta)]$$

$$= \frac{1}{4T_0} \int_{-T_0/2}^{T_0/2} [\sin^2(2\pi x) + 2 \sin(2\pi x) \sin(4\pi x) + \sin^2(4\pi x)] dx$$

HARMONICAS DIFERENTES (2π e 4π)

$$= \frac{1}{4T_0} \left[\frac{T_0}{2} + 0 + \frac{T_0}{2} \right] = \frac{1}{4T_0} \cdot T_0 = \frac{1}{4} \leadsto P = 0,25 \text{ [W]} \quad E = \infty$$

a) $E = \infty$

$$P = \frac{1}{T_0} \int_{T_0} |x(x)|^2 dx$$

$$= \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \left[\frac{1}{2} (\sin(2\pi x) + \sin(4\pi x)) \right]^2 dx$$

$$* \int_{-T_0/2}^{T_0/2} \sin(\alpha \pi x) dx = \frac{T_0}{2}$$

a) Sinal de potência

$$b) x_2(\tau) = \begin{cases} \cos(3\pi\tau) & , \quad -3 \leq \tau \leq 3 \\ 0 & , \quad \text{para} \end{cases}$$

$$E = \int_{-\infty}^{\infty} |x(\tau)|^2 d\tau$$

$$= \int_{-3}^3 \cos^2(3\pi\tau) d\tau$$

$$= \int_{-3}^3 \frac{1}{2} [1 + \cos(6\pi\tau)] d\tau = \frac{1}{2} \int_{-3}^3 1 + \cos(6\pi\tau) d\tau = \frac{1}{2} \left[\tau + \frac{\sin(6\pi\tau)}{6\pi} \right]_{-3}^3$$

$$= \frac{1}{2} \left[\left(3 + \frac{\sin(18\pi)}{6\pi} \right) - \left(-3 + \frac{\sin(-18\pi)}{6\pi} \right) \right]$$

$$= \frac{1}{2} \left[6 + \frac{\sin(18\pi)}{6\pi} - \frac{\sin(-18\pi)}{6\pi} \right] = \frac{1}{2} \left[6 + \frac{2 \sin(18\pi)}{6\pi} \right] = 3 + \frac{\sin(18\pi)}{6\pi}$$

$$E = 3 \text{ [J]} \quad \text{Sinal de energia, } P = 0$$

$$c) x_3(\tau) = \begin{cases} \tau & 0 \leq \tau \leq 2 \\ 4-\tau & 2 \leq \tau \leq 4 \\ 0 & \text{para} \end{cases}$$

$$E = \int_{-\infty}^{\infty} |x(\tau)|^2 d\tau$$

$$= \int_0^2 \tau^2 d\tau + \int_2^4 (2-\tau)^2 d\tau$$

$$= \left[\frac{\tau^3}{3} \right]_0^2 + \int_2^4 4 - 4\tau + \tau^2 d\tau = \frac{8}{3} + \left[4\tau - 2\tau^2 + \frac{\tau^3}{3} \right]_2^4$$

$$= \frac{8}{3} + \left[(16 - 32 + \frac{64}{3}) - (8 - 8 + \frac{8}{3}) \right]$$

$$= \cancel{\frac{8}{3}} + 16 - 32 + \frac{64}{3} - \cancel{8} + \cancel{8} - \cancel{\frac{8}{3}} = \frac{64}{3} - 16 = \frac{16}{3}$$

$$E = \frac{16}{3} \text{ [J]} \quad \text{Sinal de energia } P = 0$$

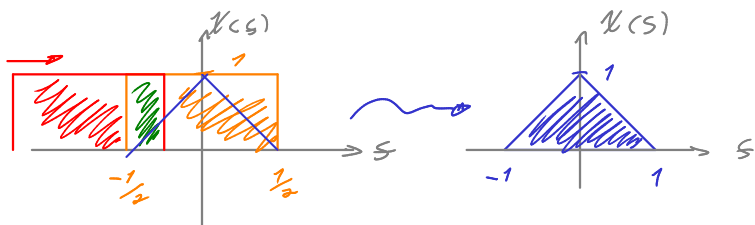
4. Determine a Transformada de Fourier dos seguintes sinais:

a) $x(t) = \text{sinc}^2(t)$;

$$x(\tau) = \text{sinc}(\tau) \cdot \text{sinc}(\tau) \Rightarrow \mathcal{F}(s) * \mathcal{F}(s)$$

$$\mathcal{F}(s) = \text{rect}(s)$$

$$\mathcal{X}(s) = \text{rect}(s) * \text{rect}(s) = \text{tri}(s)$$



$$\text{sinc}^2(\tau) \Rightarrow \text{tri}(s)$$

b) $y(t) = \frac{\text{rect}(t-3)}{41} + \frac{\text{rect}(t+3)}{42}$

$\mathcal{P}(s) = \mathcal{P}_1(s) + \mathcal{P}_2(s)$

* $\text{rect}(x) \leftrightarrow \text{sinc}(s)$

$\mathcal{P}_1(s) = \text{sinc}(s) e^{-j2\pi 3s}$

* $x(t-k) \leftrightarrow X(s) e^{-j2\pi ks}$

$\mathcal{P}_2(s) = \text{sinc}(s) e^{j2\pi 3s}$

$\mathcal{P}(s) = \text{sinc}(s) [e^{j6\pi s} + e^{-j6\pi s}]$

$\mathcal{P}(s) = \text{sinc}(s) 2 \cos(6\pi s)$

* $\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$

$2 \cos(\theta) = e^{j\theta} + e^{-j\theta}$

$\mathcal{P}(s) = 2 \cos(6\pi s) \text{sinc}(s)$

c) $z(t) = \frac{\cos^2(10\pi t)}{2} \text{sinc}(t)$

$Z(s) = Z_1(s) * Z_2(s)$

* $\cos(2\pi s) \leftrightarrow \frac{1}{2} [\delta(s-s_c) + \delta(s+s_c)]$

$Z_2(s) = \text{rect}(s)$

$z_1(t) = \cos^2(10\pi t) = \frac{1}{2} [1 + \cos(20\pi t)]$

$= \frac{1}{2} + \frac{\cos(20\pi t)}{2} \longleftrightarrow \frac{1}{2} \delta(s) + \frac{1}{2} \left[\frac{1}{2} (\delta(s-10) + \delta(s+10)) \right]$

$= \frac{1}{2} \delta(s) + \frac{1}{4} \delta(s-10) + \frac{1}{4} \delta(s+10)$

$\cos^2(10\pi t) \longleftrightarrow \frac{1}{2} \delta(s) + \frac{1}{4} \delta(s-10) + \frac{1}{4} \delta(s+10)$

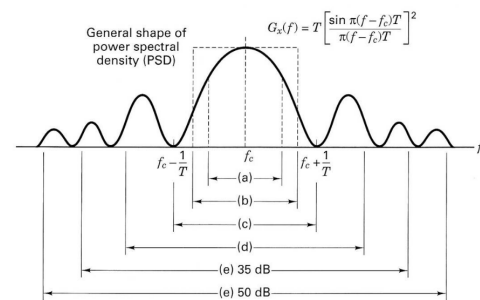
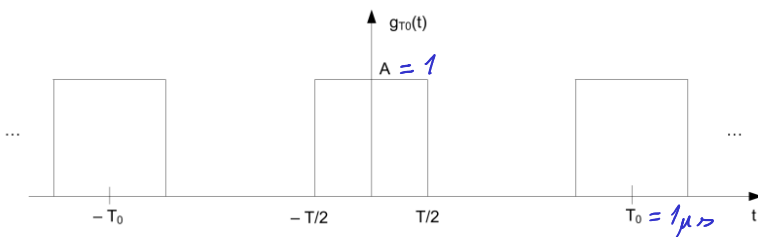
Convolução com impulso:

$y(t) = x(t) * \delta(t-k) = x(t-k)$

$Z(s) = \text{rect}(s) * \left[\frac{1}{2} \delta(s) + \frac{1}{4} \delta(s-10) + \frac{1}{4} \delta(s+10) \right]$

$Z(s) = \frac{1}{2} \text{rect}(s) + \frac{1}{4} \text{rect}(s-10) + \frac{1}{4} \text{rect}(s+10)$

5. Determine a largura de banda do trem periódico de pulsos retangulares do exemplo do slide 36 (arquivo slides_sinais_e_sistemas.pdf), pelo critério de 99% da potência. Considere $A = 1$, $T_0 = 1 \mu s$ e $T/T_0 = 0,5$.



Bandwidth of digital data. (a) Half-power. (b) Noise equivalent. (c) Null to null. (d) 99% of power. (e) Bounded PSD (defines attenuation outside bandwidth) at 35 and 50 dB.

do 99%

Calculemos C_m :

$$C_m = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} A \exp(-j2\pi m S_0 x) dx = \frac{A}{T_0} \left[-\frac{1}{j2\pi m S_0} \exp(-j2\pi m S_0 x) \right]_{-T_0/2}^{T_0/2}$$

$$= \frac{TA}{T_0} \operatorname{sinc}\left(\frac{mT}{T_0}\right) \quad \text{para } \frac{T}{T_0} = 0,5 \quad C_m = \frac{A}{2} \operatorname{sinc}\left(\frac{m}{2}\right)$$

Potência total pelo domínio do tempo:

$$P_T = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |x(x)|^2 dx = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} 1^2 dx = \frac{1}{T_0} [x]_{-T_0/2}^{T_0/2} = \frac{1}{T_0} \cdot T = 0,5 \text{ [W]}$$

$$P_T = 0,5 \text{ [W]} \quad P_{99\%} = 0,99 \cdot 0,5 \approx P_{99\%} = 0,495 \text{ [W]}$$

no Parâmetro: $P_T = \sum_{m=-\infty}^{\infty} |C_m|^2$

$$* C_m = \frac{A}{2} \operatorname{sinc}\left(\frac{m}{2}\right)$$

$$m=0 \quad C_0 = \frac{A}{2} \operatorname{sinc}\left(\frac{0}{2}\right) = \frac{A}{2} = \frac{1}{2} \quad |C_0|^2 = \frac{1}{4} = 0,25 \text{ W}$$

$$\lim_{x \rightarrow 0} \frac{\operatorname{sinc}(x)}{x} = \lim_{x \rightarrow 0} \frac{\cos(x)}{1} = 1$$

$$m=1 \quad C_1 = \frac{1}{2} \operatorname{sinc}\left(\frac{1}{2}\right) \approx 0,318 \quad 2|C_1|^2 \approx 0,2026 \text{ W} \quad |C_1| = |C_{-1}|$$

$$m=2 \quad C_2 = \frac{1}{2} \operatorname{sinc}\left(\frac{2}{2}\right) = 0 \quad 2|C_2|^2 = 0$$

$$BW \approx \frac{10}{T} (99\%)$$

$$m=3 \quad C_3 = \frac{1}{2} \operatorname{sinc}\left(\frac{3}{2}\right) \approx -0,1061 \quad 2|C_3|^2 = 0,0225$$

para m par $C_m = 0$

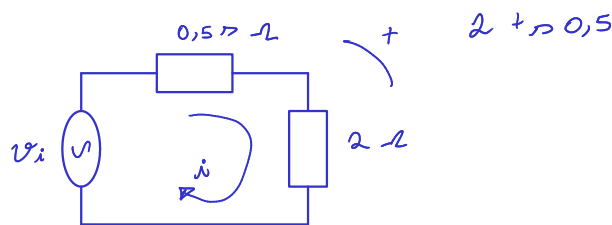
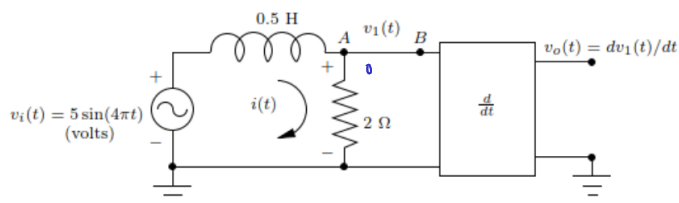
$$m=5 \quad C_5 = \frac{1}{2} \operatorname{sinc}\left(\frac{5}{2}\right) \approx 0,064 \quad 2|C_5|^2 = 0,0081$$

$$\text{calcular } 0,25 + \sum_{n=1}^{21} \left(2 \cdot \left| \frac{1}{2} \frac{\operatorname{sinc}(nx)}{nx} \right|^2 \right) \approx 0,495 = P_{99\%}$$

$$\text{Para } S_0 = 1/T_0 = 1/\mu s \sim S_0 = 1 \text{ MHz} \quad BW = m \cdot S_0 = 21 \cdot 1 \text{ M} = 21 \text{ MHz}$$

$$BW = 21 \text{ MHz}$$

6. Considere o sistema mostrado na figura a seguir. Suponha que a corrente no ramo AB é igual a zero. A tensão no ponto A é $v_1(t)$.



a) Encontre a expressão no domínio do tempo que relaciona $v_1(t)$ e $i(t)$;

$$V_i(s) = (2 + s 0,5) \cdot I(s)$$

$$v_i(x) = 2 i(x) + 0,5 \frac{d i(x)}{dt}$$

b) Utilizando a transformada de Fourier, determine a relação entre $V_1(f)$ e $V_i(f)$;

$$V_1(s) = \frac{2}{2 + s L} \cdot V_i(s) = \frac{2}{2 + j 2 \pi s \cdot 0,5} V_i(s)$$

$$V_1(s) = \frac{2}{2 + j \pi s} V_i(s) \cdot \frac{1/2}{1/2} \leadsto V_1(s) = \frac{V_i(s)}{1 + j \frac{\pi s}{2}}$$

c) Determine a relação entre $V_o(f)$ e $V_i(f)$;

$$V_o(s) = \frac{d v_1(s)}{dt} \leadsto V_o(s) = j 2 \pi s V_1(s) = j 2 \pi s \frac{V_i(s)}{1 + j \pi s \cdot \frac{1}{2}}$$

$$V_o(s) = \frac{j 2 \pi s}{1 + j \pi s \cdot \frac{1}{2}} \cdot V_i(s)$$

d) Determine $v_o(t)$;

$$V_o(s) = \frac{s}{1 + \frac{s}{4}} V_i(s) = \frac{4s}{4 + s} V_i(s)$$

$$V_o(s) = s \cdot \left(\frac{1}{4 + s} \right) \cdot 4 V_i(s) \leadsto v_o(x) = 4 \frac{d v_i(x)}{dt} * e^{-4x}$$

$$v_o(x) = 4 \int_0^x \frac{d v_i(\tau)}{d \tau} \cdot e^{-4(x-\tau)} d \tau \quad \frac{d v_i(\tau)}{d \tau} = 20 \pi \cos(4 \pi \tau)$$

$$= 80 \pi e^{-4x} \int_0^x \cos(4 \pi \tau) e^{4\tau} d \tau$$

$$\leadsto \int e^{ax} \cos(bx) dx = \frac{1}{a^2 + b^2} e^{ax} [a \cos(bx) + b \sin(bx)]$$

$$v_o(x) = 80 \pi e^{-4x} \left[\frac{e^{4\tau}}{16 + 16 \pi^2} (4 \cos(4 \pi \tau) + 4 \pi \sin(4 \pi \tau)) \right]_0^x$$

e) Determine a potência média dissipada quando a tensão de saída $v_o(t)$ é aplicada sobre uma carga de 1Ω .

$$\omega = 4\pi$$

$$V_o(s) = \frac{g \omega}{1 + g \frac{\omega}{4}} \cdot V_i(s) \rightarrow V_o(s) = \frac{g 4\pi}{1 + g\pi} V_i(s)$$

$$V_i = 5 \angle -90^\circ \text{ [V]} \quad \text{para } 5 \sin(4\pi t) \rightarrow |V_o| = \left| \frac{g 4\pi}{1 + g\pi} \right| \cdot 5 = \frac{4\pi}{\sqrt{1 + \pi^2}} \cdot 5$$

$$|V_o| = \frac{20\pi}{\sqrt{1 + \pi^2}} \text{ [V]}$$

$$V_{o,rms} = \frac{20\pi}{\sqrt{2} \sqrt{1 + \pi^2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{10\pi\sqrt{2}}{\sqrt{1 + \pi^2}} \text{ [V]}$$

$$P_{médias} = \frac{V_{o,rms}^2}{R_L} = \left(\frac{10\pi\sqrt{2}}{\sqrt{1 + \pi^2}} \right)^2 \frac{1}{R_L} = \frac{(10\pi)^2 2}{1 + \pi^2} \cdot \frac{1}{R_L}$$

$$\text{para } R_L = 1\Omega \rightarrow P_{médias} = \frac{200\pi^2}{1 + \pi^2} \rightarrow P_{médias} = 181,6 \text{ [W]}$$

7. Sejam $x(t) = e^{-t}u(t)$ e $y(t) = t^2 e^{-t}u(t)$ os sinais de entrada e saída, respectivamente, de um sistema linear e invariante no tempo (LIT). Encontre a resposta impulsiva do sistema, $h(t)$.

$$* \tau^n \longleftrightarrow \frac{n!}{s^{n+1}}$$

$$H(s) = \frac{Y(s)}{X(s)}$$

$$Y(s) = \frac{2}{(s+1)^3}$$

$$* \tau^n e^{-at} \longleftrightarrow \frac{n!}{(s+a)^{n+1}}$$

$$X(s) = \frac{1}{s+1}$$

$$* e^{-at}u(t) \longleftrightarrow \frac{1}{s+a}$$

$$H(s) = \frac{2}{(s+1)^3} \cdot (s+1) = \frac{2}{(s+1)^2} \longleftrightarrow 2 \cdot \tau e^{-\tau}u(\tau)$$

$$h(\tau) = 2\tau e^{-\tau}u(\tau)$$

8. Um sistema LIT causal e estável tem função de transferência dada por

$$H(f) = \frac{j2\pi f + 4}{6 - (2\pi f)^2 + j52\pi f}$$

$$a) \frac{g\omega + 4}{6 - \omega^2 + j5\omega}$$

a) Encontre a resposta impulsiva do sistema;

b) Determine o sinal na saída do sistema quando a entrada é dada por

$$x(t) = e^{-4t}u(t) - te^{-4t}u(t).$$

$$H(\omega) = \frac{4 + g\omega}{-\omega^2 + 5g\omega + 6}$$

$$x^2 + 5x + 6 \rightarrow x' = -2 \quad x'' = -3$$

$$H(\omega) = \frac{4 + g\omega}{(g\omega)^2 + 5g\omega + 6}$$

$$H(s) = \frac{4 + s}{(s+2)(s+3)} = \frac{A}{s+2} + \frac{B}{s+3}$$

$$4+s = A(s+3) + B(s+2)$$

$$p' \quad s=0 \quad 4 = 3A + 2B \quad \leadsto \quad 4 = 3A + 2 \cdot \left[\frac{5-4A}{3} \right]$$

$$p \quad s=1 \quad 5 = 4A + 3B \quad 4 = 3A + \frac{10}{3} - \frac{8}{3}A$$

$$\hookrightarrow 3B = 5 - 4A$$

$$B = \frac{5-4A}{3}$$

$$B = \frac{5-8}{3} = \frac{-3}{3} = -1 \therefore B = -1$$

$$4 = \frac{4}{3}A - \frac{8}{3}A + \frac{10}{3}$$

$$4 = \frac{A}{3} + \frac{10}{3}$$

$$12 = A + 10 \therefore A = 2$$

$$H(s) = \frac{2}{s+2} - \frac{1}{s+3}$$

$$* u(x) \cdot e^{-ax} \longleftrightarrow \frac{1}{s+a}$$

$$h(x) = 2 \cdot u(x) e^{-2x} - u(x) e^{-3x} \leadsto h(x) = (2e^{-2x} - e^{-3x}) u(x)$$

$$b) \quad y(x) \text{ para } x(x) = e^{-4x} u(x) - x e^{-4x} u(x)$$

$$* x^m e^{-ax} \longleftrightarrow \frac{m!}{(s+a)^{m+1}}$$

$$X(s) = \frac{1}{s+4} - \frac{1}{(s+4)^2} = \frac{s+3}{(s+4)^2}$$

$$\frac{s+4-1}{(s+4)^2}$$

$$H(s) = \frac{Y(s)}{X(s)} \therefore Y(s) = H(s) \cdot X(s)$$

$$= \frac{\cancel{4+s}}{(s+2)(\cancel{s+3})} \cdot \frac{\cancel{s+3}}{(s+4)^2} = \frac{1}{(s+2)(s+4)}$$

$$Y(s) = \frac{1}{(s+2)(s+4)} = \frac{C}{s+2} + \frac{D}{s+4}$$

$$1 = C(s+4) + D(s+2)$$

$$s=0 \quad 1 = 4C + 2D \quad \leadsto \quad 1 = 4C + 2 \left[\frac{1}{3} - \frac{5}{3}C \right]$$

$$s=1 \quad 1 = 5C + 3D$$

$$3D = 1 - 5C$$

$$1 = 4C + \frac{2}{3} - \frac{10}{3}C$$

$$D = \frac{1-5C}{3} = \frac{1}{3} - \frac{5}{3}C$$

$$1 = \frac{12C}{3} - \frac{10}{3}C + \frac{2}{3}$$

$$1 = \frac{2}{3}C + \frac{2}{3} \quad \Rightarrow \quad 3 = 2C + 2$$

$$2C = 1 \quad \therefore C = \frac{1}{2} \Rightarrow D = \frac{1 - \frac{5}{2}}{3} =$$

$$\frac{1}{3} - \frac{5}{6} = \frac{2}{6} - \frac{5}{6} = -\frac{3}{6} = -\frac{1}{2}$$

$$y(s) = \frac{1/2}{s+2} - \frac{1/2}{s+4}$$

$$* u(x) \cdot e^{-ax} \longleftrightarrow \frac{1}{s+a}$$

$$y(x) = \frac{1}{2} \cdot u(x) \cdot e^{-2x} - \frac{1}{2} u(x) e^{-4x}$$

$$y(x) = \frac{1}{2} u(x) [e^{-2x} - e^{-4x}]$$

9. Seja o processo aleatório $X(t) = A + Bt$ onde A e B são duas variáveis aleatórias independentes uniformemente distribuídas no intervalo $[-1, 1]$ Volts.

(a) Determine $\mu_X(t)$, o valor esperado do processo $X(t)$.

$$\mu_X(x) = E[X(x)]$$

$$= \int_{-\infty}^{\infty} x f_{X(x)}(x) dx$$

$$E[X(x)] = E[A + Bx]$$

$$\mu_X(x) = E[A] + E[B]x$$

Para distribuição uniforme:

$$\mu_X(x) = 0 + 0 \cdot x$$

$$E[A] = \frac{-1+1}{2} = 0 \quad E[B] = \frac{-1+1}{2} = 0$$

$$\mu_X(x) = 0$$

(b) Determine a função de autocorrelação $R_X(t_1, t_2)$.

$$R_X(x_1, x_2) = E[X(x_1) \cdot X(x_2)]$$

$$X(x_1) = A + Bx_1 \quad X(x_2) = A + Bx_2$$

$$X(x_1) \cdot X(x_2) = (A + Bx_1)(A + Bx_2) = A^2 + ABx_2 + ABx_1 + B^2x_1x_2$$

$$\begin{aligned} R_X(x_1, x_2) &= E[A^2 + AB(x_1 + x_2) + B^2x_1x_2] \\ &= E[A^2] + E[AB(x_1 + x_2)] + E[B^2]x_1x_2 \end{aligned}$$

$$\Rightarrow \text{Para } A \text{ e } B \text{ independentes } E[AB] = E[A] \cdot E[B] = 0 \cdot 0 = 0$$

$$\text{Var}(A) = E[A^2] - \underbrace{(E[A])^2}_{=0} \quad \text{Var}(A) = E[A^2]$$

$$E[A^2] = \text{Var}(A) = \frac{(1 - (-1))^2}{12} = \frac{2^2}{12} = \frac{4}{12} = \frac{1}{3}$$

$$\therefore R_X(t_1, t_2) = \frac{1}{3} + 0 \cdot (t_1 + t_2) + \frac{1}{3} t_1 t_2$$

$$R_X(t_1, t_2) = \frac{1}{3} + \frac{1}{3} t_1 t_2 \quad [V^2] \quad \text{X}$$

$$* \text{ se } t_2 = t \text{ e } t_1 = t + \tau$$

$$R_X(t + \tau, t) = E[X(t + \tau)X(t)]$$

(c) É $X(t)$ um processo estacionário no sentido amplo?

Para ser preciso $\mu_X = \text{constante}$, precisa ativar τ

$$R_X(t + \tau, t) = \frac{1}{3} (1 + (t + \tau)t) = \frac{1}{3} (1 + t^2 + \tau t) \quad \text{depende de } t, \text{ não é estacionário.}$$

11. Seja $X(t)$ um processo estacionário no sentido amplo Gaussiano de média zero e função de autocorrelação $R_X(\tau) = e^{-\pi\tau^2}$. Uma função amostra deste processo é o sinal de entrada de um sistema LIT com função de transferência dada por,

$$|H(f)| = e^{-\frac{3}{2}\pi f^2}.$$

Seja $Y(t)$ o processo estocástico que representa o sinal na saída do sistema. Determine:

$$\mu_Y(t) = E[Y(t)] = \mu_X H(0)$$

$$p/ \mu_X = E[X(t)] = 0$$

$$E[Y(t)] = 0 \cdot e^{-\frac{3}{2}\pi \cdot 0^2} = 0 \cdot 1 = 0$$

a) $E[Y(t)]; = 0$ X

b) $R_Y(\tau);$ autocorrelação da saída

$$\begin{aligned} S_X(s) &= \mathcal{F}\{R_X(\tau)\} \\ S_X(s) &= e^{-\pi s^2} \\ |H(s)|^2 &= (e^{-\frac{3}{2}\pi s^2})^2 = e^{-3\pi s^2} \\ S_{YY}(s) &= S_X(s) \cdot |H(s)|^2 = e^{-\pi s^2} \cdot e^{-3\pi s^2} = e^{-4\pi s^2} \\ R_Y(\tau) &= \mathcal{F}^{-1}\{S_{YY}(s)\} \\ R_Y(\tau) &= \frac{1}{2} e^{-\pi \tau^2} \quad \text{X} \end{aligned}$$

$* e^{-\pi s^2} \leftrightarrow e^{-\pi \tau^2}$

c) A potência média do sinal de saída do sistema.

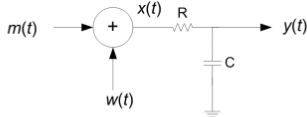
$$P_{yy} = E[y^2(t)] = R_{yy}(0)$$

$$R_{yy}(0) = \frac{1}{2} e^{-\frac{\pi \cdot 0^2}{4}} \rightarrow R_{yy}(0) = \frac{1}{2} e^{-\frac{\pi \cdot 0^2}{4}} = \frac{1}{2} \cdot 1 = \frac{1}{2} \text{ [W]}$$

12. O sinal $m(t) = 5 \cos(2000\pi t)$ é corrompido por ruído branco aditivo Gaussiano. A versão corrompida deste sinal é dada por:

$$x(t) = m(t) + w(t),$$

onde $w(t)$ é uma função amostra de um processo de ruído branco Gaussiano $W(t)$ de média zero e densidade espectral de potência $\frac{N_0}{2} = 10^{-9}$ Watts/Hz. O sinal $x(t)$ é passado através de um sistema LIT produzindo na saída o sinal $y(t)$ como mostrado na figura.



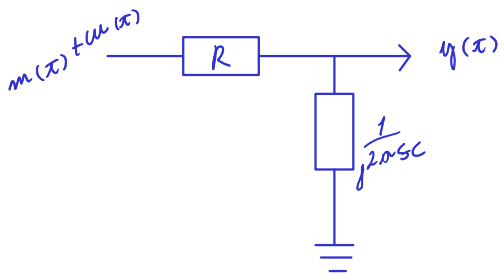
$$m(t) = 5 \cos(2000\pi t)$$

$$x(t) = m(t) + w(t)$$

$$f_0 = \frac{2000\pi}{2\pi} = 1 \text{ kHz}$$

$$P_m = \frac{5^2}{2} = 12,5 \text{ [W]}$$

Determine a relação sinal-ruído, SNR, na saída do sistema linear, isto é, determine a razão entre a potência da componente de sinal de $y(t)$ e a potência de ruído contida em $y(t)$ supondo $R = 300\Omega$ e $C = 1 \mu\text{F}$. Expresse o resultado em dB.



$$y(s) = \frac{\frac{1}{j2\pi s C}}{R + \frac{1}{j2\pi s C}} \quad \text{(m(s) + w(s))}$$

$$\frac{y(s)}{x(s)} = \frac{1}{j2\pi s C R + 1} = H(s)$$

Potência sinal de saída:

$$P_{mo} = P_m \cdot |H(f_0)|^2$$

$$|H(1\text{kHz})|^2 \approx \frac{1}{1 + j2\pi \cdot 1\text{kHz} \cdot 300 \cdot 1\mu\text{F}} \approx 0,2196$$

$$P_{mo} = 12,5 \cdot 0,2196 \approx 2,745 \text{ [W]}$$

Potência do ruído na saída (P_{wo}):

$$P_{wo} = \int_{-\infty}^{\infty} \frac{N_0}{2} |H(s)|^2 ds = \frac{N_0}{2} \int_{-\infty}^{\infty} \frac{1}{1 + (2\pi s RC)^2} ds = \frac{N_0}{2} \cdot \frac{1}{2RC}$$

$$P_{wo} = 10^{-9} \cdot \frac{1}{2 \cdot 300 \cdot 10^{-6}} = 10^{-9} \cdot \frac{1}{6 \cdot 10^{-4}} = \frac{10^{-5}}{6} \approx 1,667 \cdot 10^{-6} \text{ [W]}$$

$$\text{SNR} = \frac{P_{mo}}{P_{wo}} = \frac{2,745}{1,667 \cdot 10^{-6}} \approx 1,647 \cdot 10^6 \quad \text{SNR}_{\text{dB}} = 10 \log_{10}(1,647 \cdot 10^6) \approx 62,17 \text{ dB}$$

$$\text{SNR}_{\text{dB}} \approx 62,17 \text{ [dB]}$$